

<i>Element</i>	<i>Mass No.</i>	<i>Atomic Mass (AMU)</i>	<i>Element</i>	<i>Mass No.</i>	<i>Atomic Mass (AMU)</i>
⁶³ Eu	152	151.919756	⁷⁶ Os	187 ^a	186.955833
	154	153.922282		184	183.952750
	151	150.919838		186	185.953870
	153	152.921242		187	186.955832
⁶⁴ Gd	152	151.919794	⁷⁷ Ir	188	187.956081
	154	153.920929		189	188.958300
	155	154.922664		190	189.958630
	156	155.922175		192	191.961450
⁶⁵ Tb	157	156.924025	⁷⁸ Pt	191	190.960640
	158	157.924178		193	192.963012
	160	159.927115		190 ^a	189.959950
	159	158.925351		192	191.961150
⁶⁶ Dy	156	155.923930	⁸² Pb	194	193.962725
	158	157.924449		195	194.964813
	196	195.964967		205	204.974442
	198	197.967895		204	203.973044
⁷⁹ Au	197	196.966541	⁸³ Bi	206	205.974468
⁸⁰ Hg	196	195.965820		207	206.975903
	198	197.966756		208	207.976650
	199	198.968279		209	208.981082
⁸¹ Tl	200	199.968327	⁹⁰ Th	232 ^a	232.038124
	201	200.970308	⁹² U	234 ^a	234.040904
	202	201.970642		235 ^a	235.043915
	204	203.973495		238 ^a	238.050770
	203	202.972353			

^aUnstable nuclide.
Source: From Weidner, R., and Sells, R. *Elementary Modern Physics*, 2nd edition. Boston, Allyn & Bacon, 1968. Used by permission.

ANSWERS TO SELECTED PROBLEMS

Chapter 2

- 2-1. 8; 16; 0.1369 amu; 127.5 MeV; 8.0 MeV/nucleon
2-2. 15.999 amu
2-4. Tungsten, 58, 220 eV; hydrogen, 10.1 eV

- 2-5. 0.51 MeV
2-6. 2.37×10^{24} , 0.85 g
2-7. 15.2×10^{15} J; 3.63×10^{12} kcal

Chapter 3

- 3-1. 4.6 hr; 6.9 hr; 156 hr
3-2. 3.5×10^{-7} ; 6.6×10^{15} ; 1.8×10^{-6}
3-4. 36.7%
3-6. $2.4 \times 10^{19} \text{ sec}^{-1}$; 0.124 Å
3-8. 3.14 hr
3-10. 6.15×10^{14} ; 9.2×10^{-8} g

- 3-11. 30.3 mCi
3-12. 1.09 days; 4100 Ci; 16.4 Ci
3-13. ^{74}As , positron decay and/or electron capture;
 ^{76}As , negatron decay
3-14. 80

Chapter 4

- 4-1. 2.031 keV/cm
4-2. 51,699 IP/cm
4-4. 0.59
4-5. 0.94 cm

- 4-7. 12 keV; 35 keV
4-9. 138 keV; 12 keV; decreased
4-10. 865 keV

Chapter 5

- 5-2. 3.12×10^{17} ; 5000 J/sec
5-4. 11.5 degrees
5-5. No; yes
5-6. No; 1 should be eliminated

- 5-7. 3 per min
5-8. 250 KeV; 0.023; 0.05 A
5-9. 2 mm
5-10. 9.6 mm

Chapter 6

- 6-1. $1.06 \times 10^9 \text{ MeV}/(\text{m}^2\text{-sec})$; $1.06 \times 10^{10} \text{ MeV}/\text{m}^2$
6-2. 0.68 J/kg
6-3. $363 \text{ J}/\text{m}^2$; $2.27 \times 10^{15} \text{ photons}/\text{m}^2$
6-4. 2.4×10^{11}
6-5. 7.4
6-6. $1.53 \times 10^{-10} \text{ A}$
6-7. 52 picofarads

- 6-8. 270 R
6-11. $10^{-4} \text{ J}/\text{g}$; 10^{-3} J
6-12. 0.85
6-13. 1,500 mrem
6-14. 0.014°C
6-15. 1.44×10^{18}
6-17. (a) 1.5 mm Al; (b) 2.25 mm Al; (c) 0.7

Chapter 8

- 8-1. (a) No; (b) yes, yes; (c) no
8-2. No

- 8-4. 4.25×10^6
8-5. 0.6 mV

Chapter 9

- 9-1. 3.6×10^{10}
 9-2. 10,500; 2000
 9-3. 1.60; 60%
 9-6. (a) 75 mCi/mL; (b) 0.064
 9-9. 2.76 MeV = photopeak; 2.25 MeV = single-escape peak;
 1.74 MeV = double-escape peak; 1.38 MeV = photopeak;
 0.87 MeV = single-escape peak; 0.51 MeV = annihilation peak;
 0.20 MeV = backscatter peak
 9-11. About 580
 9-14. 220 MCi
 9-15. 3400 yr
 9-16. 2.84×10^5 Ci/g

Chapter 10

- 10-1. 5120 bits = 5×2^{10}
 10-2. 503,316,480 bits = $8 \times 60 \times 2^{20}$
 62,914,560 bytes = 60×2^{20}
 10-3. 16,777,216 bits = $32 \times 0.5 \times 2^{20}$
 2,097,152 bytes = $4 \times 0.5 \times 2^{20}$
 10-4. 20,480 images
 10-5. 31.25 mV
 10-6. PROM
 10-7. 1.8 (line pairs)/mm < 2.5 (line pairs)/mm, does not meet the standard
 10-8. 4.8 Mbit

Chapter 11

- 11-1. a. 6 cpm, 1.67%
 b. 4.7 cpm, 3.60%
 c. 8 cpm, 3.32%
 11-2. a. The precision is described by the 2-mm standard deviation. Accuracy relates the researcher's data to the "true" answer determined by some independent method, which in this case is the study of excised bone. The degree of accuracy is therefore given by the 2% difference between the means of the two methods of bone measurement. The results are biased because the original measurements are consistently larger than the true value.
 b. Both mean and standard deviation are "scaled" by the same proportion (see Table 11-3). Corrected mean = $(0.98)(5.5) = 5.4$ cm. Corrected standard deviation = $(0.98)(2 \text{ mm}) = 1.96 \text{ mm} = 0.02$ cm.
Note: The corrected values are given with the same number of significant figures as the original data, and the same units (cm) are used for both.
 11-3. t value = 2.31; $P = 0.022$

Chapter 12

- 12-1. 59%
 12-2. 5500 mL
 12-4. 43%
 12-5. (a) ^{241}Am ; (b) ^{197}Hg

Chapter 13

- 13-1. 10% and 3%
 13-2. 126.5 mR

Chapter 14

- 14-1. 3240
 14-3. Yes
 14-4. 9 MHz
 14-6. 1.8×10^{-3} c/(kg-min), 7.0 roentgen/min
 7.1×10^{-3} c/(kg-min), 27.7 roentgen/min

Chapter 16

- 16-1. Unsharpness, contrast, noise, distortion and artifacts
 16-2. Geometric, subject, motion, receptor, unsharpness
 16-3. 0.4 mm
 16-4. 1.8 mm
 16-5. 24 cm
 16-6. Subject, technique, contrast agents, receptor
 16-7. Structure noise, radiation noise, receptor noise, quantum mottle

Chapter 17

- 17-1. 0.2 mm; 2.5 line pairs/mm
 17-2. Higher; projected focal spot is smaller
 17-3. 0.41

Chapter 18

- 18-2. The individual sees at 20 ft what an ordinary person sees at 100 ft.
 18-4. Sensitivity = $p(TP) = TP/(TP + FN)$
 Specificity = $p(TN) = TN/(TN + FP)$
 18-6. Specificity, because a false-negative means that a tumor that should have been detected is missed.

Chapter 19

- 19-2. -13 dB
 19-3. (a) $\alpha_R = 0.011$ $\alpha_T = 0.989$
 (b) $\alpha_R = 0.011$ $\alpha_T = 0.989$
 19-4. 33.4 degrees
 19-6. 6.9 dB

Chapter 20

- 20-2. 1.2 mm
 20-4. 48 cm

Chapter 21

- 21-2. 128
 21-3. $\frac{1}{4}$
 21-5. Attenuation

Chapter 22

- 22-1. 325 Hz; higher
 22-2. 1.3 kHz
 22-5. 650 Hz; a, decrease; b, increase

Chapter 23

- 23-2. 7.63×10^{18}
 23-3. 1.98 tesla
 23-4. 1/16 of the maximum signal

Chapter 24

- 24-1. 11.52 min
 24-2. a
 24-3. 280 Hz

Chapter 28

- 28-1. 30 rem
 28-2. 300 mrem
 28-3. 440 mR
 28-4. 2.0 mm Pb
 28-5. None
 28-6. 0.4 mm Pb
 28-7. 0.7 mm Pb
 28-8. 1.27 mm Pb

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